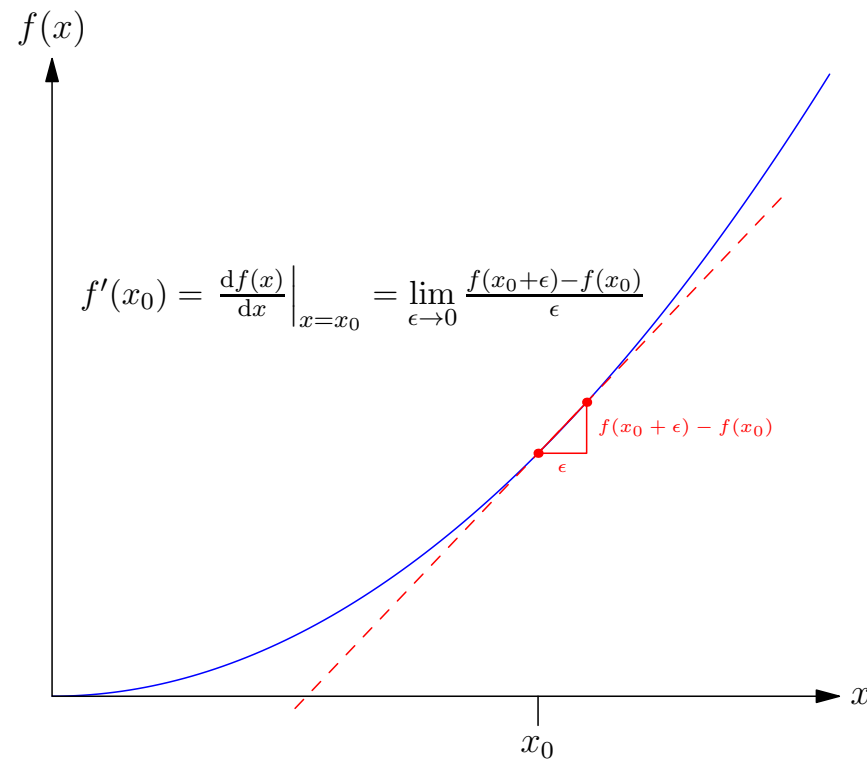


# Mathematics of Machine Learning

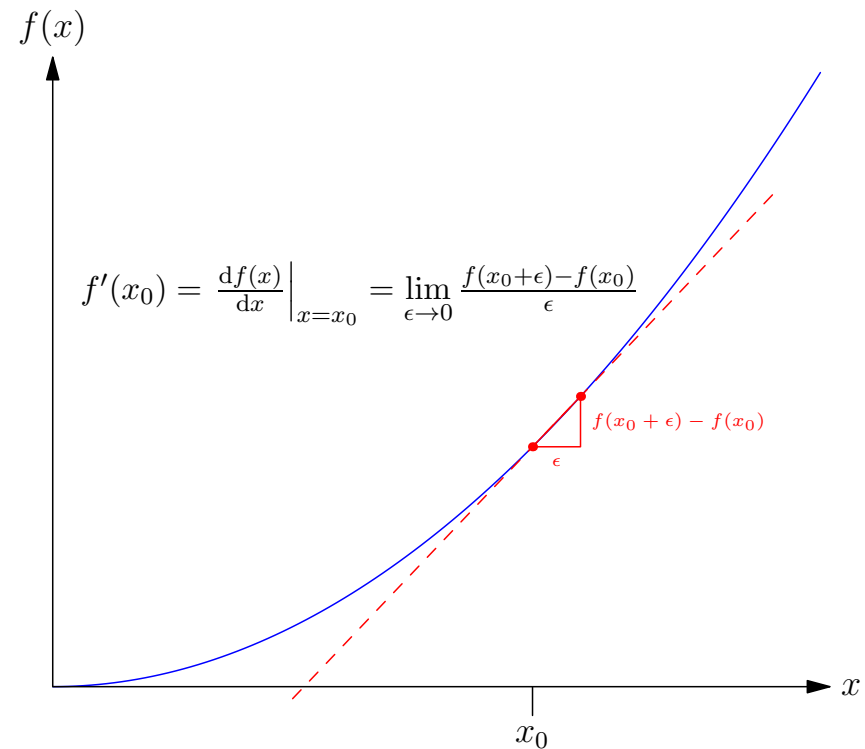
## *Differential Calculus*



*Differentiation, product and chain rules, vectors and matrices*

# Outline

1. **Why Calculus?**
2. Differentiation
3. Vector Calculus
4. Matrix Calculus



# Why Calculus?

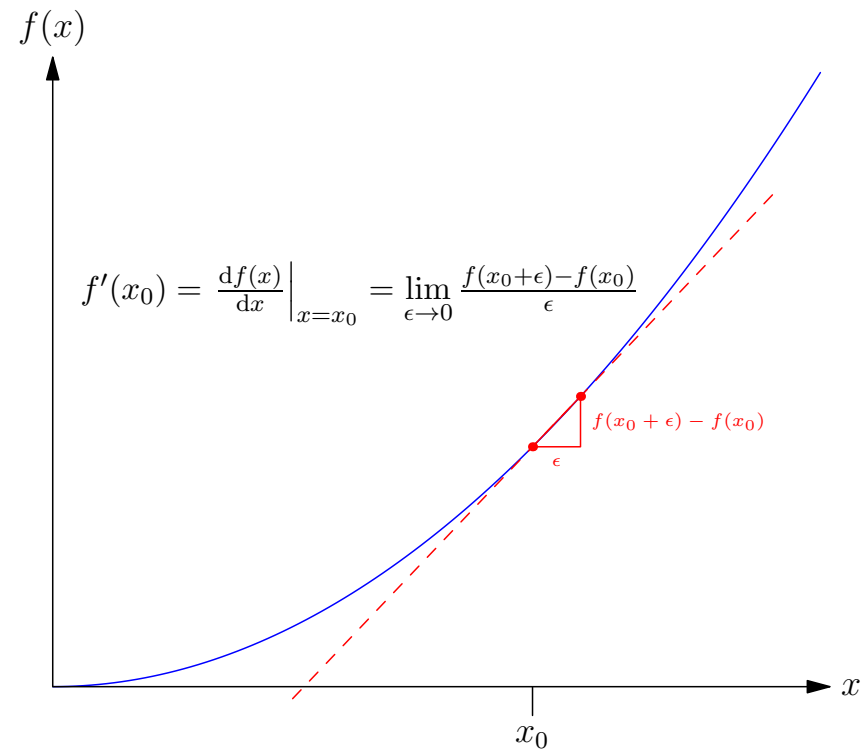
- Calculus is a fundamental tool of mathematical analysis■
- In machine learning differentiation is fundamental tool in optimisation■
- Integration is an essential tool in taking expectations over continuous distributions■
- Both differentiation and integration crop up elsewhere■

# Back to Basics

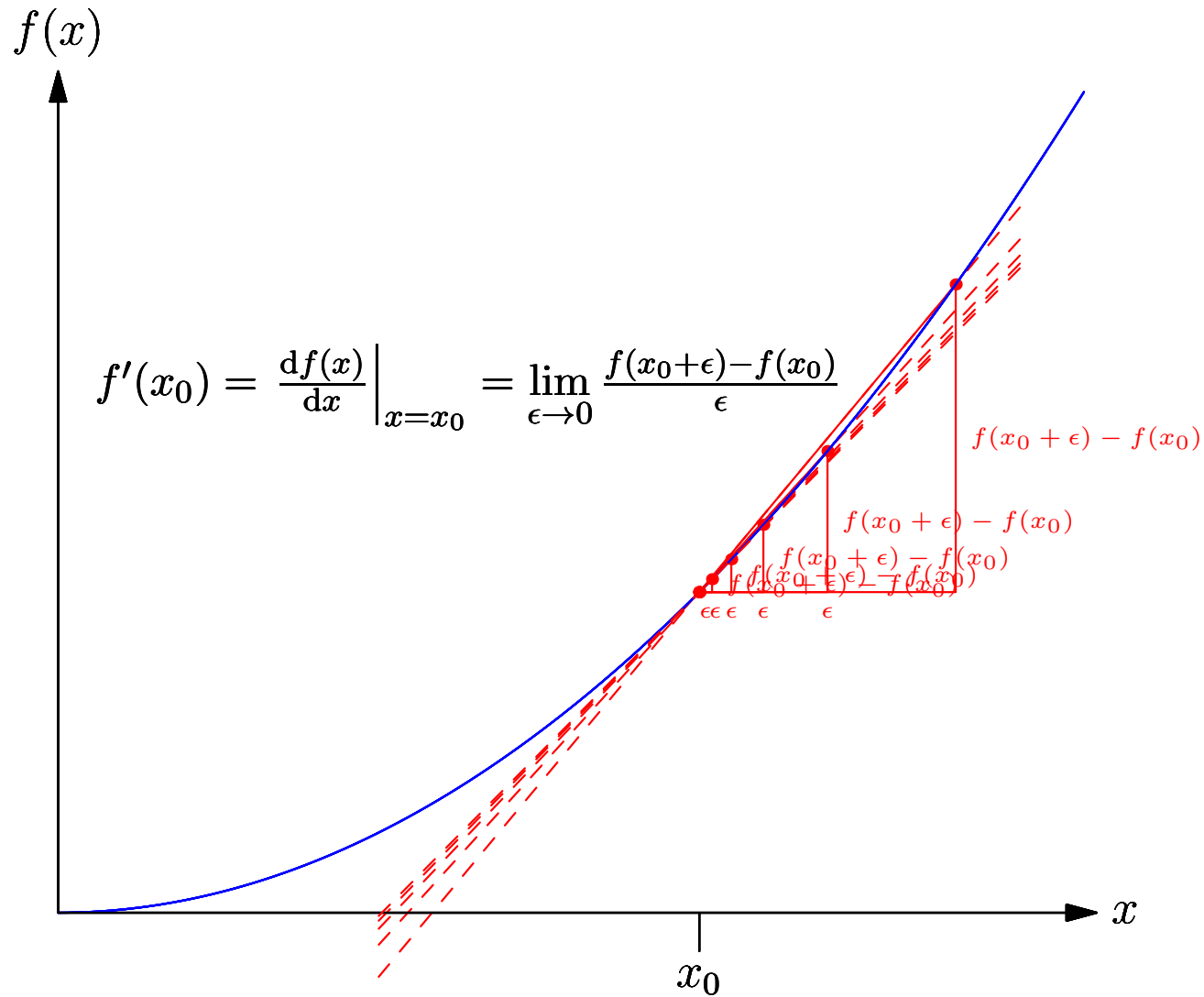
- You have all done A-level maths so should be familiar with the rules of calculus■
- But, it is easy to forget the rules and sometimes we use quite sophisticated tricks■
- Although the sophisticated tricks really speed up calculations, it pays to be able to understand where these tricks come from■

# Outline

1. Why Calculus?
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# Differentiation



# Polynomials

- $f(x) = x^2$

$$\begin{aligned}\frac{dx^2}{dx} &= \lim_{\epsilon \rightarrow 0} \frac{(x + \epsilon)^2 - x^2}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{(x^2 + 2\epsilon x + \epsilon^2) - x^2}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} 2x + \epsilon = 2x\end{aligned}$$

- $(x + \epsilon)^n = (x + \epsilon)(x + \epsilon) \cdots (x + \epsilon) = x^n + n\epsilon x^{n-1} + O(\epsilon^2)$

$$\frac{dx^n}{dx} = \lim_{\epsilon \rightarrow 0} \frac{(x + \epsilon)^n - x^n}{\epsilon} = \lim_{\epsilon \rightarrow 0} n x^{n-1} + O(\epsilon) = n x^{n-1}$$

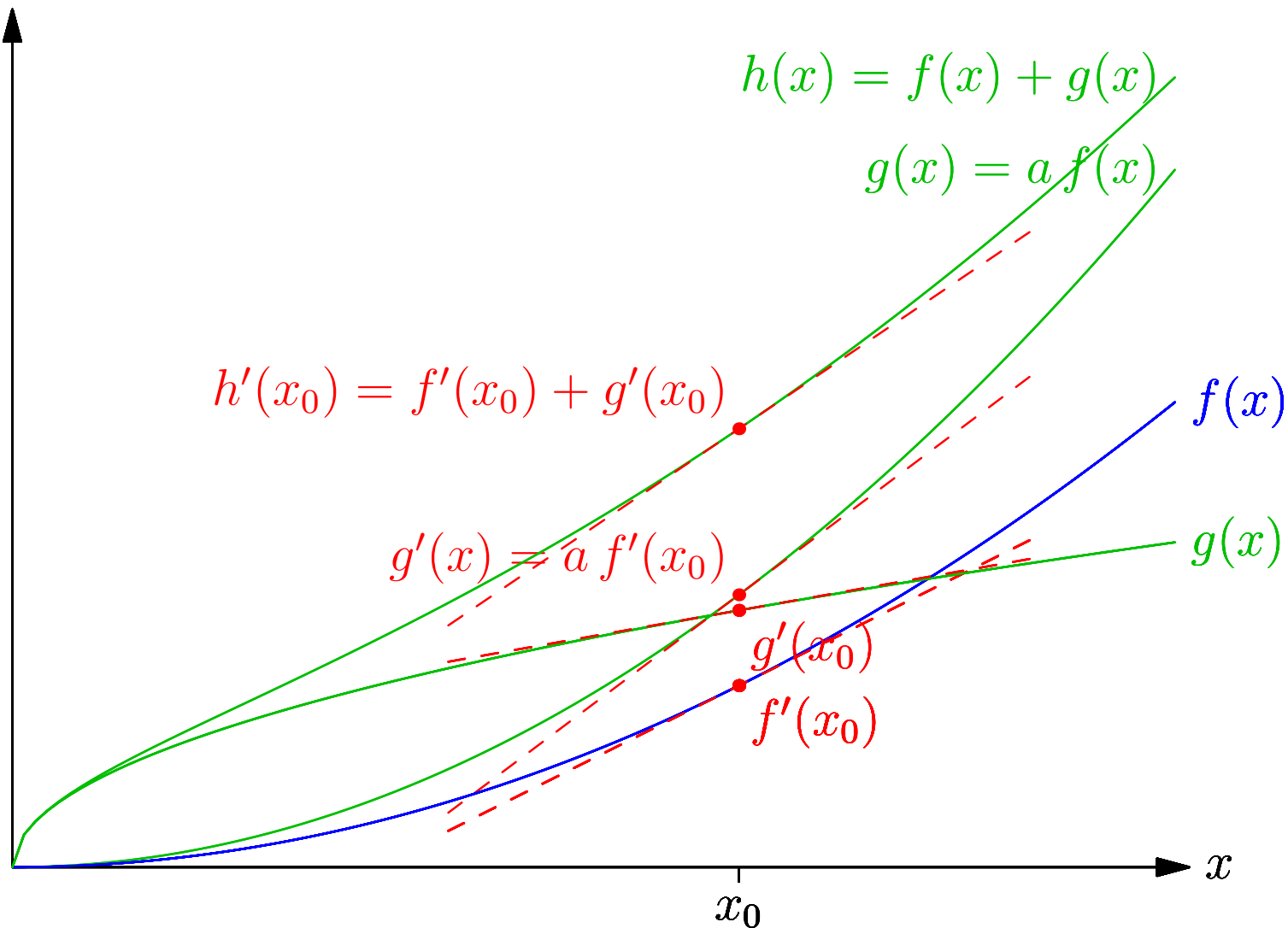
# Linearity of derivatives

- Consider the derivative of  $a f(x) + b g(x)$

$$\begin{aligned}\frac{d(a f(x) + b g(x))}{dx} &= \lim_{\epsilon \rightarrow 0} \frac{(a f(x + \epsilon) + b g(x + \epsilon)) - (a f(x) + b g(x))}{\epsilon} \\&= \lim_{\epsilon \rightarrow 0} \frac{a(f(x + \epsilon) - f(x)) + b(g(x + \epsilon) - g(x))}{\epsilon} \\&= \lim_{\epsilon \rightarrow 0} a \left( \frac{f(x + \epsilon) - f(x)}{\epsilon} \right) + b \left( \frac{g(x + \epsilon) - g(x)}{\epsilon} \right) \\&= a f'(x) + b g'(x)\end{aligned}$$

- Differentiation is a linear operation!**

# Linearity in Pictures



# Product Rule

- Recall  $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- If  $h(x) = f(x)g(x)$

$$\begin{aligned} h'(x) &= \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon)g(x + \epsilon) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{(f(x) + \epsilon f'(x) + O(\epsilon^2))(g(x) + \epsilon g'(x) + O(\epsilon^2)) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\epsilon(f'(x)g(x) + f(x)g'(x)) + O(\epsilon^2)}{\epsilon} = f'(x)g(x) + f(x)g'(x) \end{aligned}$$

- This is the **product rule**

# Chain Rule

- Recall  $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- Let  $h(x) = f(g(x))$ ■
- Then

$$\begin{aligned} h(x + \epsilon) &= f(g(x + \epsilon)) \text{■} = f(g(x) + \epsilon g'(x) + O(\epsilon^2)) \text{■} \\ &= f(g(x)) + \epsilon g'(x) f'(g(x)) + O(\epsilon^2) \text{■} \end{aligned}$$

- Thus

$$h'(x) = \lim_{\epsilon \rightarrow 0} \frac{h(x + \epsilon) - h(x)}{\epsilon} = g'(x) f'(g(x)) \text{■}$$

- This is the famous **chain rule**■ Together with the product rule it means you can differentiate almost everything■

# More on chain rules

- We can also write the chain rule as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

- Sometimes this is neater or easier to remember

$$\begin{aligned} \frac{de^{\cos(x^2)}}{dx} &= \frac{de^{\cos(x^2)}}{d\cos(x^2)} \frac{d\cos(x^2)}{dx^2} \frac{dx^2}{dx} \\ &= e^{\cos(x^2)} (-\sin(x^2)) 2x \\ &= -2x \sin(x^2) e^{\cos(x^2)} \end{aligned}$$

# Inverse functions

- Suppose  $g(y) = f^{-1}(y)$  is the inverse of  $f(x)$  in the sense that  $g(f(x)) = f^{-1}(f(x)) = x$
- Using the chain rule

$$\frac{dg(f(x))}{dx} = f'(x)g'(f(x)) = 1$$

since  $g(f(x)) = x$

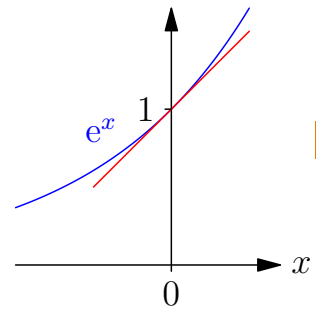
- So  $g'(f(x)) = 1/f'(x)$
- Writing  $y = f(x)$  so that  $x = f^{-1}(y) = g(y)$  we find  $g'(y) = 1/f'(g(y))$  that is

$$\frac{dg(y)}{dy} = \frac{1}{f'(g(y))} \qquad \frac{df^{-1}(y)}{dy} = \frac{1}{f'(f^{-1}(y))}$$

# Exponentials

- Note that  $a^{b+c} = a^b a^c$  (that is we multiply  $a$  together  $b + c$  times)■

- Now  $e^\epsilon \approx (1 + \epsilon)$



- But  $e^{x+\epsilon} = e^x e^\epsilon = e^x (1 + \epsilon + O(\epsilon^2))$ ■  $= e^x + \epsilon e^x + O(\epsilon^2)$ ■

$$\frac{de^x}{dx} = \lim_{\epsilon \rightarrow 0} \frac{e^{x+\epsilon} - e^x}{\epsilon} \text{■} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon e^x + O(\epsilon^2)}{\epsilon} = e^x \text{■}$$

# Functions of Exponentials

- What about  $f(x) = e^{cx}$

$$\frac{de^{cx}}{dx} = \frac{de^{cx}}{dcx} \frac{dcx}{dx} = ce^{cx}$$

- More generally using the chain rule

$$\frac{de^{g(x)}}{dx} = g'(x)e^{g(x)}$$

- Also  $a^{bc} = (a^b)^c$  (that is we multiply  $a$  together  $b \times c$  times)

$$\frac{da^x}{dx} = \frac{d(e^{\ln(a)})^x}{dx} = \frac{de^{\ln(a)x}}{dx} = \ln(a)e^{\ln(a)x} = \ln(a)a^x$$

# Natural Logarithms

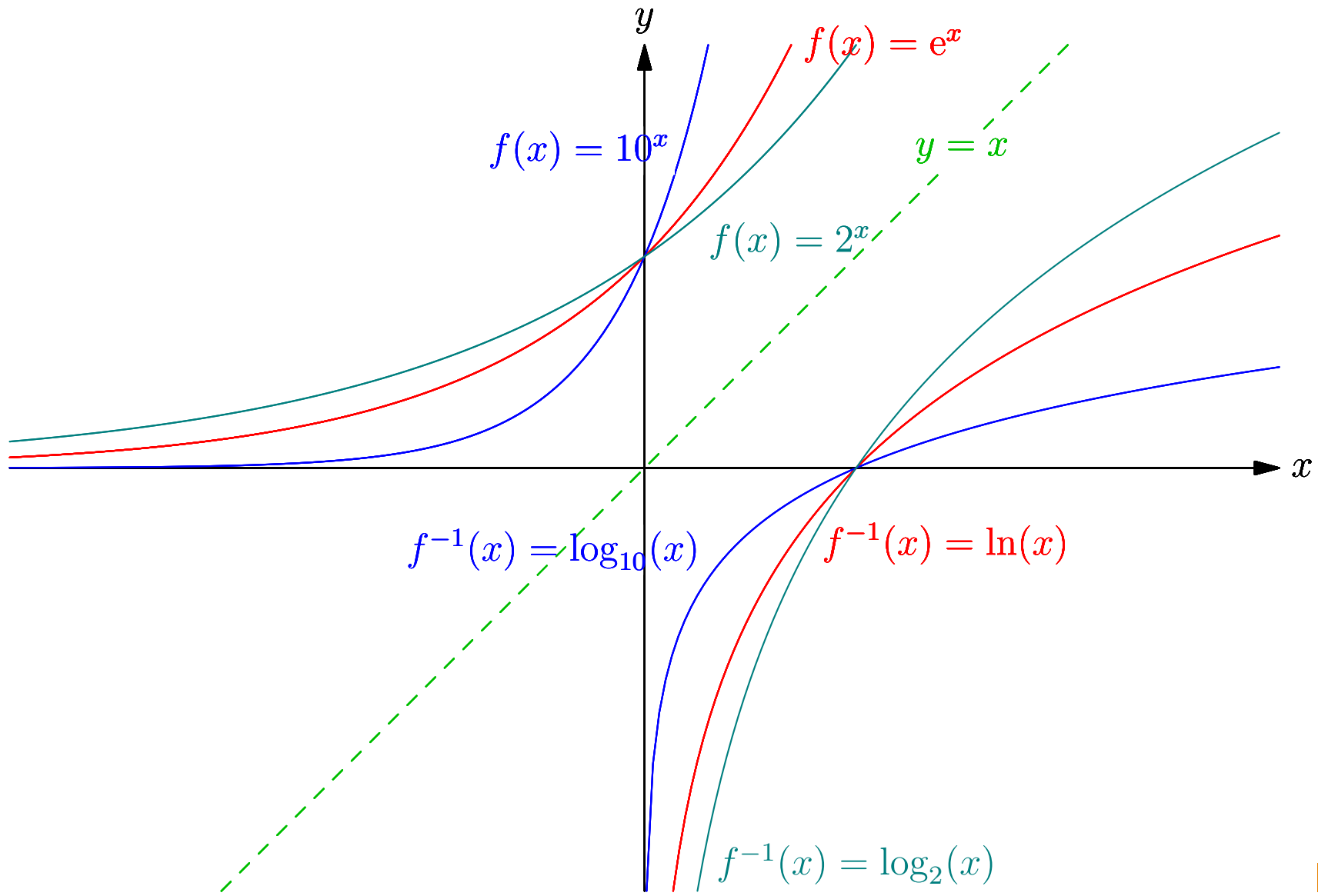
- The natural logarithm is defined as the inverse of  $e^x$

$$\ln(e^x) = x \qquad e^{\ln(y)} = y$$

- Recall that if  $g(y) = f^{-1}(y)$  then  $g'(y) = 1/f'(g(y))$
- Consider  $g(y) = \ln(y)$  and  $f(x) = e^x$  (with  $f'(x) = e^x$ )

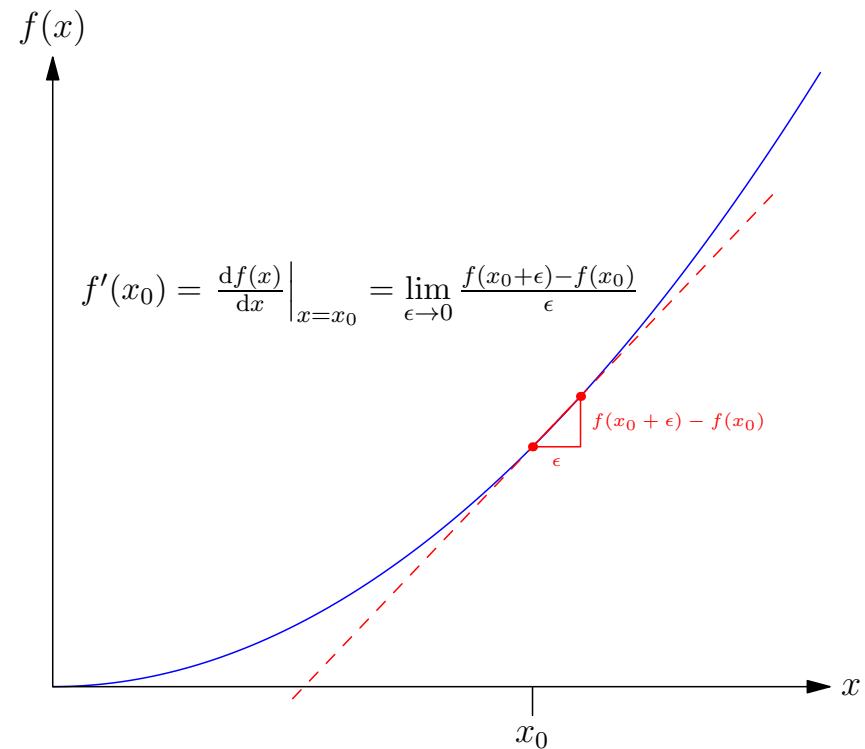
$$\frac{d\ln(y)}{dy} = \frac{1}{e^{\ln(y)}} = \frac{1}{y}$$

# Exponentials and Logarithms



# Outline

1. Why Calculus?
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# Derivatives in High Dimensions

- When working with functions  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  in many dimensions then there will typically be different derivative in different directions■
- To compute the derivative in a direction  $\mathbf{u} \in \mathbb{R}^n$  (where  $\|\mathbf{u}\| = 1$ ) at a point  $\mathbf{x} \in \mathbb{R}^n$  we use

$$\partial_{\mathbf{u}} F(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon} \blacksquare$$

- If  $\mathbf{u} = \boldsymbol{\delta}_i = (0, \dots, 0, 1, 0, \dots, 0)$  (i.e.  $u_i = 1$ ) then

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \boldsymbol{\delta}_i) - f(\mathbf{x})}{\epsilon} \blacksquare$$

# Taylor

- If we expand  $f(\mathbf{x} + \epsilon \mathbf{u})$  to first order in  $\epsilon$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + O(\epsilon^2)$$

then  $g_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}$  ■

- Recall we defined the vector of first order derivatives of  $f(\mathbf{x})$  to be the gradient

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix} \quad \blacksquare$$

- Thus

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2) \quad \blacksquare$$

This is the start of the high-dimensional Taylor expansion ■

# Computing Gradients 1

- We can compute the gradient by writing out  $f(\mathbf{x})$  componentwise and performing the partial derivative with respect to  $x_i$

$$\begin{aligned}\nabla \mathbf{w}^\top \mathbf{M} \mathbf{w} &= \begin{pmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \frac{\partial}{\partial w_3} \\ \vdots \end{pmatrix} \sum_{i,j} w_i M_{ij} w_j = \begin{pmatrix} \sum_j M_{1j} w_j + \sum_i w_i M_{i1} \\ \sum_j M_{2j} w_j + \sum_i w_i M_{i2} \\ \sum_j M_{3j} w_j + \sum_i w_i M_{i3} \\ \vdots \end{pmatrix} \\ &= \mathbf{M} \mathbf{w} + \mathbf{M}^\top \mathbf{w} = (\mathbf{M} + \mathbf{M}^\top) \mathbf{w}\end{aligned}$$

- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

# Computing Gradients 2

- A slicker way is just to expand  $f(x + \epsilon u)$ ■

- Consider  $f(x) = x^\top M x + a^\top x$

$$\begin{aligned} f(x + \epsilon u) &= (x + \epsilon u)^\top M (x + \epsilon u) + a^\top (x + \epsilon u) \blacksquare \\ &= f(x) + \epsilon (u^\top M x + x^\top M u + a^\top u) + O(\epsilon^2) \blacksquare \\ &= f(x) + \epsilon u^\top (M x + M^\top x + a) + O(\epsilon^2) \end{aligned}$$

using  $x^\top M u = u^\top M^\top x$  and  $a^\top u = u^\top a$ ■

- But  $f(x + \epsilon u) = f(x) + \epsilon u^\top \nabla f(x) + O(\epsilon^2)$  so

$$\nabla f(x) = M x + M^\top x + a \blacksquare$$

# Function Composition

- The gradient tells us about the change in a function value when we perturb an input  $\mathbf{x}$  by a small amount in the directions  $\mathbf{u}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2) \blacksquare$$

- What happens when we compose functions:  $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$   $\blacksquare$   
Note that,

$$\mathbf{x} \in \mathbb{R}^n \quad \mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad f : \mathbb{R}^m \rightarrow \mathbb{R} \quad h : \mathbb{R}^n \rightarrow \mathbb{R} \blacksquare$$

- The vector valued function  $\mathbf{g}(\mathbf{x})$  can be seen as  $m$  independent

$$\text{function } \mathbf{g}(\mathbf{x}) = \begin{pmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{pmatrix} \blacksquare$$

# Jacobians

- Perturbing  $\mathbf{x}$  by  $\epsilon \mathbf{u}$

$$g_i(\mathbf{x} + \epsilon \mathbf{u}) = g_i(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla g_i(\mathbf{x}) + O(\epsilon^2)$$

so that

$$\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u}) = \mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u} + O(\epsilon^2)$$

where  $\mathbf{J}$  is an  $m \times n$  matrix with components  $J_{ij} = \frac{\partial g_i(\mathbf{x})}{\partial x_j}$

- Note that  $\mathbf{J} \mathbf{u}$  is a column vector of size  $m$
- Ignoring terms of order  $\epsilon^2$

$$\begin{aligned} h(\mathbf{x} + \epsilon \mathbf{u}) &= f(\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u})) \\ &= f(\mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u}) \\ &= f(\mathbf{g}(\mathbf{x})) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \end{aligned}$$

# Chain Rule in High Dimensions

- Since

$$h(\mathbf{x} + \epsilon \mathbf{u}) = h(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla_{\mathbf{x}} h(\mathbf{x}) + O(\epsilon^2) \blacksquare$$

and (recall  $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$ )

$$h(\mathbf{x} + \epsilon \mathbf{u}) = h(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \blacksquare$$

- We get that

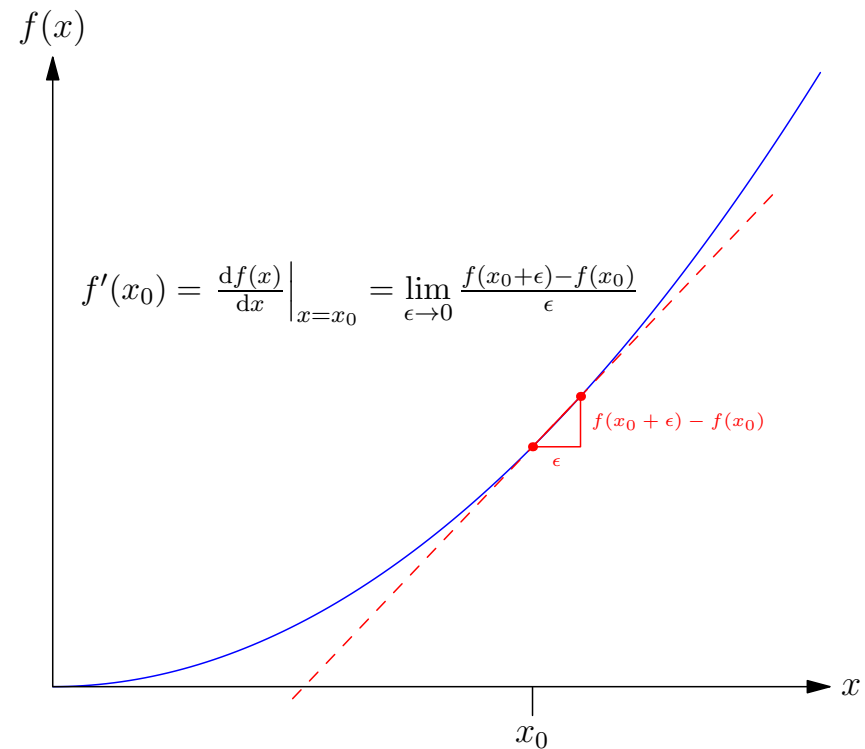
$$\nabla_{\mathbf{x}} h(\mathbf{x}) = \nabla_{\mathbf{x}} f(\mathbf{g}(\mathbf{x})) = \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \blacksquare$$

- Componentwise

$$\frac{\partial h(\mathbf{x})}{\partial x_i} = \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial x_i} \blacksquare = \sum_{j=1}^m \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial g_j} \frac{\partial g_j(\mathbf{x})}{\partial x_i} \blacksquare$$

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# Differentiating Matrices

- Often we have loss functions with respect to a matrix  $\mathbf{W}$ , e.g.

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

- We might want to find the minimum with respect to  $\mathbf{W}$
- This occurs at a point  $\mathbf{W}^*$  where  $L(\mathbf{W})$  does not increase as we change  $\mathbf{W}$  in any way
- That is, we seek a  $\mathbf{W}^*$  such that, for any matrices  $\mathbf{U}$

$$L(\mathbf{W}^* + \epsilon \mathbf{U}) - L(\mathbf{W}^*) = O(\epsilon^2)$$

# Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial L(\mathbf{W})}{\partial W_{11}} & \frac{\partial L(\mathbf{W})}{\partial W_{12}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{1m}} \\ \frac{\partial L(\mathbf{W})}{\partial W_{21}} & \frac{\partial L(\mathbf{W})}{\partial W_{22}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(\mathbf{W})}{\partial W_{n1}} & \frac{\partial L(\mathbf{W})}{\partial W_{n2}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{nm}} \end{pmatrix} \quad \blacksquare$$

- From an identical argument we used for vectors

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \text{tr} \mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} + O(\epsilon^2) \quad \blacksquare$$

where

$$\text{tr} \mathbf{U}^\top \mathbf{G} = \sum_i [\mathbf{U}^\top \mathbf{G}]_{ii} = \sum_{ij} U_{ji} G_{ji} = \sum_{ij} U_{ij} G_{ij} = \langle \mathbf{U}, \mathbf{G} \rangle \quad \blacksquare$$

# Example

- Suppose

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

then

$$\begin{aligned} L(\mathbf{W} + \epsilon \mathbf{U}) &= (\mathbf{a}^\top (\mathbf{W} + \epsilon \mathbf{U}) \mathbf{b} - c)^2 = (\mathbf{a}^\top \mathbf{W} \mathbf{b} + \epsilon \mathbf{a}^\top \mathbf{U} \mathbf{b} - c)^2 \\ &= L(\mathbf{W}) + 2\epsilon (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) (\mathbf{a}^\top \mathbf{U} \mathbf{b}) + O(\epsilon^2) \end{aligned}$$

- Now

$$\mathbf{a}^\top \mathbf{U} \mathbf{b} = \sum_{ij} a_i U_{ij} b_j = \sum_{ij} U_{ji} a_j b_i = \text{tr} \mathbf{U}^\top \mathbf{a} \mathbf{b}^\top$$

$$\text{Thus } \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = 2 (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a} \mathbf{b}^\top$$

# Traces

- The trace of a matrix is the sum of its diagonal elements

$$\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T = \sum_i A_{ii}$$

- Clearly  $\text{tr} c\mathbf{A} = c \text{tr} \mathbf{A}$
- Also  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$
- We note that

$$\text{tr} \mathbf{A} \mathbf{B} = \sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} B_{ij} A_{ji} = \text{tr} \mathbf{B} \mathbf{A}$$

- It follows that

$$\text{tr} \mathbf{A} \mathbf{B} \mathbf{C} \mathbf{D} = \text{tr} \mathbf{D} \mathbf{A} \mathbf{B} \mathbf{C} = \text{tr} \mathbf{C} \mathbf{D} \mathbf{A} \mathbf{B} = \text{tr} \mathbf{B} \mathbf{C} \mathbf{D} \mathbf{A}$$

# Quick Matrix Differentiation

- Let

$$\partial_{\mathbf{U}} f(\mathbf{X}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{X} + \epsilon \mathbf{U}) - f(\mathbf{X})}{\epsilon} = \text{tr } \mathbf{U}^\top \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}$$

- E.g.

$$\begin{aligned} \partial_{\mathbf{U}} \text{tr } \mathbf{A} \mathbf{X} \mathbf{B} &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{tr } \mathbf{A} (\mathbf{X} + \epsilon \mathbf{U}) \mathbf{B} - \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}) \\ &= \text{tr } \mathbf{A} \mathbf{U} \mathbf{B} = \text{tr } \mathbf{B}^\top \mathbf{U}^\top \mathbf{A}^\top = \text{tr } \mathbf{U}^\top \mathbf{A}^\top \mathbf{B}^\top \end{aligned}$$

thus

$$\frac{\partial \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}}{\partial \mathbf{X}} = \mathbf{A}^\top \mathbf{B}^\top$$

# Summary

- With care you can differentiate most expressions■
- The chain and product rule are incredibly powerful tools■
- We can generalise differentiation to vectors and matrices■
- There are a number of surprisingly useful results■ see **The Matrix Cookbook**■
- When we look at **integration** it gets harder■